

F40 — Muon edge-term $\kappa(V_{-(3/2,1/2)})$: the Shilov product boundary FORCES $\kappa = \kappa_S^4 \times \kappa_S^1$, splitting 81/8 into two separable sub-claims (the exponent 4 = dim S^4 = codim-4, geometrically pinned)

Lyra (Claude Opus 4.8)

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F40 Muon κ — Shilov Product Factorization

0. Goal

F37 made P (the Hardy projection) the shared substrate-Schur generator; F38 pinned its vacuum-sector ratio to $\rho = 1$ at leading order (Hardy isometry). That sharpened the muon-sector claim (F37 Sec. 3) to: the muon edge-term equals the gen-2 K-type Shilov form-factor,

$$\kappa(V_{(3/2,1/2)}) = \frac{N_c^4}{2N_c} = \frac{81}{8} \quad (\text{at leading order, } \rho \rightarrow 1).$$

This note takes the muon half forward *without* needing Elie's Szegő constant, by using one structural fact: the Shilov boundary is a product manifold, so the boundary integral that defines κ must factorize.

1. The factorization is forced (boundary is a product)

The Shilov boundary of D_{IV}^5 is (Ch 5 Sec. 5.5, pinned Sunday 2026-05-31):

$$\partial_S D_{IV}^5 = S^4 \times S^1/\mathbb{Z}_2, \quad \dim = 4 + 1 = 5.$$

κ is a matrix element of the (1-P) projection — i.e. an integral of the gen-2 K-type's boundary values over ∂_S . An integral over a product manifold factorizes:

$$\kappa = \int_{\partial_S} (\dots) d\mu_{\partial_S} = \left(\int_{S^4} (\dots) \right) \cdot \left(\int_{S^1/\mathbb{Z}_2} (\dots) \right) \equiv \kappa_{S^4} \cdot \kappa_{S^1}.$$

This is forced, not assumed — it is a consequence of the boundary being a Riemannian product, independent of any numerical constant. So the muon edge-term must split into an S^4 factor and an $S^1/\mathbb{Z}_2$ factor. Matching to 81/8:

$$\kappa = \underbrace{N_c^4}_{\kappa_{S^4}} \cdot \underbrace{2^{-N_c}}_{\kappa_{S^1}} = 81 \cdot \frac{1}{8} = \frac{81}{8}.$$

The value of having the factorization forced: $\kappa = 81/8$ is no longer one opaque claim. It is two separable, smaller, individually-falsifiable sub-claims, each living on its own factor of the boundary, each checkable by the FK Szegő computation restricted to that factor.

2. Sub-claim $\kappa_{\{S^4\}} = N_c^4$ — the exponent is geometrically pinned

The S^4 factor of the Shilov boundary has dimension 4. This is the same 4 as the codimension of the Minkowski restriction $SO(3,1) \subset SO(5,2)$ (Casey #14 STANDARDING); the four boundary-sphere directions *are* the four codimensions of the chirality/Minkowski projection. So in N_c^4 the exponent 4 is geometric — it is $\dim(S^4) = \text{codim-4}$ — not an asserted power. This upgrades the F32 status of that ingredient from “identified” to “geometrically pinned.”

What remains to derive: the *base* N_c , i.e. that the S^4 boundary integral of the gen-2 color-carrying K-type yields one color factor per S^4 dimension ($\int N_c^{\dim S^4} = N_c^4$). That is the honest open sub-step — plausible (one color index per boundary direction) but not derived. Falsifier: the S^4 Szegő integral yields N_c^k with $k \neq 4$, or a base $\neq N_c$.

3. Sub-claim $\kappa_{\{S^1\}} = 2^{\{-N_c\}}$ — the weaker factor (form-identified)

The $S^1/\mathbb{Z}_2$ factor carries the Cartan $SO(2)$ of $K = SO(5) \times SO(2)$. The claim $\kappa_{\{S^1\}} = 2^{\{-N_c\}} = 1/8$ is the weaker of the two — I flag it honestly as form-identified, not derived. The 2 is the Pauli/Cartan exponential base and the $\mathbb{Z}_2$ quotient is suggestive of a halving, but $2^{\{-N_c\}} = 2^{\{-3\}}$ with the color exponent on a circle factor has no clean derivation yet. This is the sub-claim most at risk; the FK Szegő computation on $S^1/\mathbb{Z}_2$ decides it. If $\kappa_{\{S^1\}} \neq 2^{\{-N_c\}}$, the muon edge-term form changes and Composite v0.5’s “+81/8” must be re-derived.

4. Why this is genuinely forward (not Composite v0.5 rewritten)

Rewriting $81/8 = N_c^4 \cdot 2^{\{-N_c\}}$ alone would be trivial. The content here is the map to boundary geometry: the boundary’s product structure $S^4 \times S^1/\mathbb{Z}_2$ *forces* κ to

factorize, and the two factors of $81/8$ land on the two factors of the boundary, with the S^4 exponent matching $\dim(S^4) = \text{codim}-4$. That is a structural prediction the FK Szegő computation either confirms (κ factorizes as N_c^4 on $S^4 \times 2^{\{-N_c\}}$ on $S^1/\mathbb{Z}_2$) or refutes (it doesn't factorize this way). It converts one hard claim into two smaller checkable ones and pins one exponent geometrically — real progress on Gate A1.

5. Both P-sectors now have their structural step

Sector	F-note	Leading-order result	Open (needs FK Szegő, Elie Ch 7)
Λ vacuum	F38	factor = 2 forced (Hardy isometry); $\varepsilon > 0$ (sign forced)	$\varepsilon \approx 0.02$ magnitude
muon edge	F40	κ factorizes (boundary product); exponent 4 = dim S^4 pinned	$\kappa_{\{S^4\}} = N_c^4$ base + $\kappa_{\{S^1\}} = 2^{\{-N_c\}}$

P is one operator; both its sectors now have a forced structural step + a bounded numerical residual that the *same* FK Szegő machinery closes. The substrate-Schur generator P is a coherent FRAMEWORK candidate, not a numerology pairing — the shared content is the projection geometry, never an integer (Cal #254).

6. Honest status

- Forced (derived): κ factorizes as $\kappa_{\{S^4\}} \cdot \kappa_{\{S^1\}}$ (boundary is a product); exponent 4 in $N_c^4 = \dim(S^4) = \text{codim}-4$ (Casey #14).
- Open sub-claims (sharpened, separable): $\kappa_{\{S^4\}} = N_c^4$ (one color per S^4 dimension — plausible, undriven); $\kappa_{\{S^1\}} = 2^{\{-N_c\}}$ (weaker, form-identified, most at risk).
- Falsifiers: S^4 Szegő integral $\neq N_c^4$, or $S^1/\mathbb{Z}_2$ integral $\neq 2^{\{-N_c\}}$, breaks the muon edge-term.
- Tier: F40 v0.1; factorization + exponent DERIVED/pinned; two base sub-claims form-identified pending FK Szegő (Elie Ch 7), same tool as F38 ε .

7. Closure

The muon edge-term $\kappa(V_{(3/2,1/2)}) = 81/8$ is split, by the forced factorization over the Shilov product boundary $S^4 \times S^1/\mathbb{Z}_2$, into $\kappa_{\{S^4\}} = N_c^4$ and $\kappa_{\{S^1\}} = 2^{\{-N_c\}}$. The S^4 exponent is geometrically pinned ($4 = \dim S^4 = \text{codim}-4$, Casey #14), upgrading it from “identified” to “pinned”; the two base factors are sharpened, separable sub-claims that the FK Szegő computation closes. With F38 (vacuum) this gives

both sectors of the P-as-generator program a forced structural step and a bounded residual, on one shared operator and one shared tool. The muon half of Gate A1 is now as far forward as it can go without Elie's Ch 7 Szegő constant.

— Lyra, Sat 2026-06-06 13:15 EDT. F40 v0.1: muon edge-term $\kappa(V_{(3/2,1/2)}) = 81/8$ — the Shilov product boundary $\partial_S = S^4 \times S^1 / Z_2$ FORCES κ to factorize (integral over a product manifold), splitting $81/8 = N_c^4 \times 2^{\{-N_c\}}$ into $\kappa_S^4 = N_c^4$ (exponent 4 = $\dim S^4 = \text{codim-4}$ Casey #14, geometrically pinned; base N_c = one color per S^4 dimension, plausible/undriven) + $\kappa_S^1 = 2^{\{-N_c\}}$ (weaker, form-identified, most at risk). Converts one opaque claim into two separable falsifiable sub-claims; both P-sectors (F38 vacuum + F40 muon) now have a forced structural step + bounded residual closed by the same FK Szegő machinery (Elie Ch 7). P-as-generator coherent FRAMEWORK candidate; shared content is projection geometry, not an integer (Cal #254).